

Two-Body Problem & Kepler Solver

Thu-Phuong Duong

National Taiwan Normal University, Taipei 116, Taiwan

Abstract—This is an assignment report of the “Celestial Mechanics” course given by Asst. Prof. David M. Hernandez at National Taiwan Normal University. This report aims to investigate the numerical solution to the two-body orbital problem using the incremental Kepler equation and Gauss (f and g) coefficients. A Python-based numerical solver was developed to propagate the motion of a body orbiting a solar-mass star. The Newton-Raphson, Bisection, and Secant method are implemented to solve the transcendental Kepler equation. The simulation’s physical validity was assessed by monitoring the conservation of total mechanical energy and angular momentum. Results indicate that the f and g propagation method maintains high stability and accuracy, with relative errors in conserved quantities approaching 10^{-16} . The influence of time-step size (Δt) and orbital eccentricity (e) on convergence and numerical stability was also explored, confirming the robustness of the algorithm under varying orbital conditions.

I. INTRODUCTION

The motion of two celestial bodies under the influence of their mutual gravitational attraction represents the most fundamental problem in celestial mechanics. Known as the Classical Two-Body Problem, its analytical foundation was established by Johannes Kepler’s empirical laws and later generalized by Isaac Newton’s Universal Law of Gravitation. While the relative motion of such a system is constrained to a conic section—specifically an ellipse for bound orbits—the mapping of this geometric path to a precise temporal evolution remains a non-trivial computational challenge (Murray & Dermott, 1999).

A. Theoretical Framework

The dynamics of two point masses m_0 and m_1 are dictated by Newton’s Second Law and the Law of Universal Gravitation. By defining the relative position vector $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_0$, the equation of motion is reduced to a single vector differential equation:

$$\ddot{\mathbf{r}} + \frac{\mu}{r^3} \mathbf{r} = 0$$

where $\mu = G(m_0 + m_1)$ represents the gravitational parameter. In this assignment, I adopt a canonical unit system where $G = 4\pi^2 \text{ AU}^3 \text{ yr}^{-2} M_\odot^{-1}$, effectively normalizing the orbital period of a test particle at 1 AU to unity.

For bound systems ($E < 0$), the resulting trajectory is constrained to an ellipse characterized by the semi-major axis a and eccentricity e . In these systems, the geometry of the orbit is coupled to time through Kepler’s Laws. While Kepler’s second law dictates that a radius vector joins any planet to the Sun and sweeps out equal areas in equal lengths of time, the mathematical challenge is determining the exact position at a

specific time. This requires the use of the Mean Anomaly (M) and the Eccentric Anomaly (E). The mapping between these domains is expressed by the transcendental Kepler Equation, $M = E - e \sin E$.

B. Incremental Kepler Equation

A critical component of this assignment is the propagation of motion over discrete time intervals Δt . Rather than solving for the total anomaly from a fixed epoch, modern computational efficiency is gained by solving the incremental Kepler equation. This equation relates the change in mean anomaly (ΔM) to the change in eccentric anomaly (ΔE) over a timestep:

$$\Delta M = \Delta E + (1 - \cos \Delta E) e \sin E - (\sin \Delta E) e \cos E$$

By solving for the displacement ΔE rather than the absolute anomaly, the propagator maintains higher precision across varying scales of Δt . This equation forms the core numerical challenge, requiring iterative solvers to bridge the gap between mean temporal progress and eccentric angular displacement.

C. Gauss Coefficients

Once ΔE is resolved, the system’s state must be updated. Instead of integrating the equations of motion numerically (which can lead to significant truncation errors), this report aims to utilize the Gauss coefficients ($f, g, \dot{f}, \dot{g}$). These coefficients provide a closed-form solution to propagate the position and velocity vectors forward in time:

$$\vec{r}_{t+\Delta t} = f \vec{r}_t + g \vec{v}_t$$

$$\vec{v}_{t+\Delta t} = \dot{f} \vec{r}_t + \dot{g} \vec{v}_t$$

This approach is analytically exact within the framework of the Keplerian problem, as it constrains the state update to the invariant orbital plane, thereby preserving the integrals of motion to the limits of machine precision.

D. Objective and Scope

The objective of this report is to develop a numerical propagator for a two-body system. Specifically, I will evaluate the performance and stability of the f and g coefficient method across varying orbital geometries and temporal resolutions. The scope of this investigation includes developing a solver for the incremental Kepler equation using Newton-Raphson, Bisection, and Secant methods; and investigating the impact of the time step (Δt) and eccentricity (e) on the convergence of the root-finding algorithms; and assessing the accuracy of the propagator by monitoring the relative errors in total

mechanical energy and angular momentum, aiming for numerical precision near the machine epsilon ($\sim 10^{-16}$); and systematically exploring how orbital shape and mass ratios influence the behavior and stability of the numerical approach.

II. METHODOLOGY

A. The Discrete State-Space Propagator

The propagation of a two-body system over a finite temporal interval Δt can be expressed as a linear transformation of the initial state vector $\mathbf{X}_0 = [\mathbf{r}_0, \mathbf{v}_0]^T \in \mathbb{R}^6$. I employ the Lagrangian Coefficients ($f, g, \dot{f}, \dot{g}$), which provide an analytical mapping based on the geometry of the conic section:

$$\mathbf{r}_{n+1} = f\mathbf{r}_n + g\mathbf{v}_n$$

$$\mathbf{v}_{n+1} = \dot{f}\mathbf{r}_n + \dot{g}\mathbf{v}_n$$

Mathematically, this approach ensures that the state propagation occurs along the Hamiltonian flow of the Keplerian potential. By constraining the transformation to the orbital plane, the semi-major axis a , eccentricity e , and inclination i are preserved as integrals of motion, effectively eliminating the orbital drift common in non-symplectic numerical methods.

B. The Incremental Kepler Equation

The transition between states at t_n and t_{n+1} is driven by the Mean Motion $n = \sqrt{\mu/a^3}$, where the linear progress of time is expressed through the change in Mean Anomaly:

$$\Delta M = n(t_{n+1} - t_n) = n\Delta t$$

The core challenge lies in mapping this linear temporal progress ΔM to the non-linear angular progress ΔE . The Incremental Kepler Equation is then derived by taking the difference of the classical Kepler equation at two states and applying trigonometric sum-to-product identities:

$$\Delta M = \Delta E + e \sin E_n (1 - \cos \Delta E) - e \cos E_n \sin \Delta E$$

This incremental formulation is numerically superior to solving the absolute equation $M = E - e \sin E$. By isolating the angular displacement ΔE , I mitigate cancellation errors that typically plague high-precision simulations when $\Delta t \ll$ the orbital period.

C. Numerical Root-Finding Methods

Given that ΔE cannot be isolated algebraically, I implement and compare three iterative root-finding architectures to solve the residual function $f(\Delta E) = 0$.

The Newton-Raphson method leverages the analytical Jacobian of the Keplerian residual to achieve quadratic convergence ($\epsilon_{n+1} \approx \epsilon_n^2$) and is defined as

$$f'(\Delta E) = 1 + e \sin E_n \sin \Delta E - e \cos E_n \cos \Delta E$$

The update follows $\Delta E_{k+1} = \Delta E_k - f(\Delta E_k)/f'(\Delta E_k)$.

The Secant method approximates $f'(\Delta E)$ using two previous points and is useful when the analytical derivative is computationally expensive

$$\Delta E_{k+1} = \Delta E_k - f(\Delta E_k) \frac{\Delta E_k - \Delta E_{k-1}}{f(\Delta E_k) - f(\Delta E_{k-1})}$$

The Bisection method provides guaranteed, albeit linear, convergence. It serves as a robust benchmark for validating the convergence basins of the more aggressive gradient-based solvers. It requires bracketing the root within $[a, b]$ such that $f(a) \cdot f(b) < 0$.

D. Implementation of the Gauss Coefficients

Once ΔE is resolved to a tolerance $\epsilon \leq 10^{-16}$, the Lagrangian coefficients are computed as follows:

$$f = 1 - \frac{a}{r_n} (1 - \cos \Delta E)$$

$$g = \Delta t - \frac{1}{n} (\Delta E - \sin \Delta E)$$

$$\dot{f} = -\frac{\sqrt{\mu a}}{r_{n+1} r_n} \sin \Delta E$$

$$\dot{g} = 1 - \frac{a}{r_{n+1}} (1 - \cos \Delta E)$$

Here, the magnitude of the future radius r_{n+1} is determined by the updated eccentric anomaly: $r_{n+1} = a(1 - e \cos(E_n + \Delta E))$. This analytical update ensures that the system's specific energy and angular momentum vectors are conserved to the numerical precision of the underlying floating-point architecture.

III. RESULTS AND DISCUSSION

A. Baseline Simulation

Table I summarizes the initial conditions and numerical constraints utilized for the baseline simulation. These parameters correspond to a Jupiter-mass object around a solar-mass primary. By changing the parameters in this table, I can explore sensitivity of the model in relation to different parameters.

TABLE I
BASELINE SIMULATION PARAMETERS

Parameter	Symbol	Value	Unit
Central Mass	m_0	1.0	$M_\odot$
Orbiting Mass	m_1	0.001	$M_\odot$
Gravitational Parameter	μ	$4\pi^2$	AU^3/yr^2
Semi-major Axis	a	1.0	AU
Eccentricity	e	0.2	Dimensionless
Time Step	Δt (or dt)	0.2	Years
Convergence Tolerance	τ	10^{-16}	Dimensionless

The baseline model with the initial settings is illustrated in Figure 1, presenting a significant departure from ideal Keplerian motion. The state transition mapping ensures a closed loop orbit. However, the resulting trajectory is notably non-elliptical, and the conservation of Hamiltonian invariants shows fluctuations and a trend of increasing over time. More

specifically, the energy and angular momentum errors are around 10^{-3} and 10^{-13} respectively, which is much higher than the 10^{-16} limit. These errors spike when the planet is closest to the Sun, where it moves the fastest. The growing trend is the result of big time step.

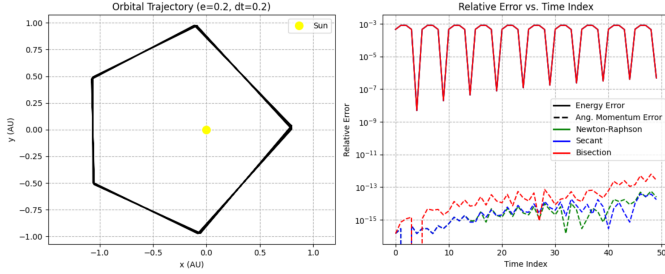


Fig. 1. The left figure shows elliptical orbital trajectory of a two-body system with eccentricity 0.2, propagated with a timestep of $\Delta t = 0.2$ years, the Sun is located at the origin (0,0) in the (x, y) plane. The right figure shows the semi-log plot of relative error of the simulation over time using 3 different numerical root-finding methods.

B. Influence of Eccentricity on Orbital Geometry

Figure 2 illustrates the transition of the orbital manifold as the eccentricity (e) varies from 0.1 to 0.9. In all cases, the primary mass (the Sun) is fixed at the origin (0,0). As the eccentricity increases, the periapsis distance ($r_p = a(1 - e)$) decreases, causing the orbiting body to pass significantly closer to the Sun. At higher eccentricities, the apoapsis distance increases, resulting in the stretched elliptical shapes observed in the $e = 0.9$ (bronze) and $e = 0.8$ (grey) trajectories. The polygonal appearance remains consistent across all cases because the time step is held at $\Delta t = 0.2$ years. However, a critical numerical observation is seen at high eccentricity orbits: the spacing between calculated points becomes extremely tight near periapsis and very wide near apoapsis. This is a visual manifestation of Kepler's Second Law, the body must cover a much larger angular distance in the same amount of time (Δt) when it is closest to the gravitational center. For $e = 0.9$, the sharp "spike" near the origin suggests that the numerical solver is under extreme stress, as the velocity changes rapidly over a single time step.

According to the Vis-viva equation:

$$v^2 = \mu \left(\frac{2}{r} - \frac{1}{a} \right)$$

Since a is constant for all these orbits, they all have the same total energy, yet their geometries vary wildly. Figure 2 demonstrates that eccentricity determines the shape and angular momentum of the system, while the semi-major axis alone dictates the energy and orbital period.

C. Effect of Timestep on Numerical Accuracy

Figure 3 demonstrates the system behavior when the temporal resolution is increased by a factor of 20 ($\Delta t = 0.01$ years). Geometrically, the trajectory now appears as a smooth ellipse, confirming that the previous polygonal shapes were

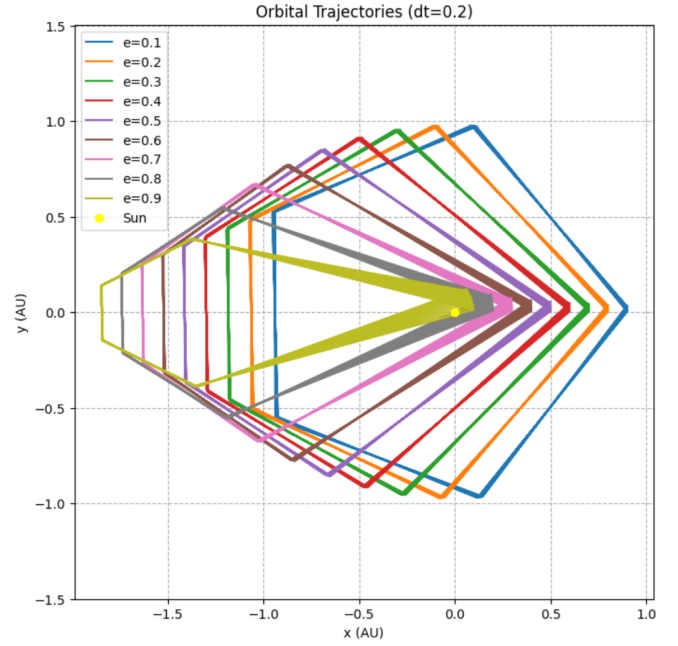


Fig. 2. Elliptical orbital trajectory of a two-body system with different eccentricities ranging from 0.1 to 0.9, propagated with a timestep of $\Delta t = 0.2$ years, the Sun is located at the origin (0,0) in the (x, y) plane.

merely an artifact of undersampling. By reducing the time step, the visual representation aligns with the expected Keplerian geometry.

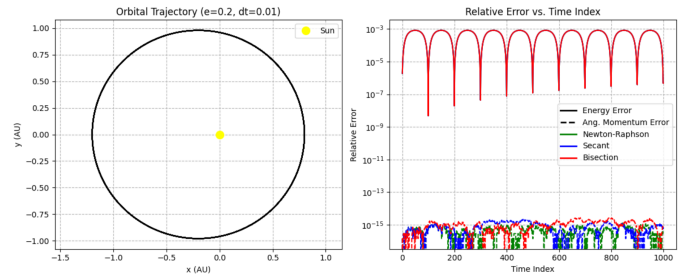


Fig. 3. Orbital trajectory and error convergence for $e = 0.2$ at high temporal resolution ($\Delta t = 0.01$ years).

The right panel reveals a fascinating error analysis between energy and angular momentum conservation. The relative energy error remains high, oscillating between 10^{-3} and 10^{-5} . While the Newton-Raphson and Secant methods overlap, the Bisection method occasionally reaches higher precision near 10^{-8} . The periodic peaks correlate with periapsis, where the orbital velocity is at its maximum. The fact that the energy error did not drop significantly with Δt suggests that it is likely limited by the precision of the transcendental solver's tolerance rather than the timestep itself. In stark contrast, the relative angular momentum error has successfully dropped to the 10^{-15} level, effectively reaching the limit precision. In the f and g formulation, the angular momentum is often more stable because the cross product $\mathbf{r} \times \mathbf{v}$ is less sensitive to the specific resolution of the eccentric anomaly than the scalar energy equation, which depends on the inverse of the

radius ($1/r$).

Figure 4 presents the model's behavior under a highly aggressive temporal discretization of $\Delta t = 0.3$ years. This timestep value results in fewer than four samples per revolution. The left panel shows a star-shaped trajectory, which is a classic example of aliasing in celestial mechanics. Because the sampling rate is so low, the linear interpolation between points fails to capture the curvature of the orbit.

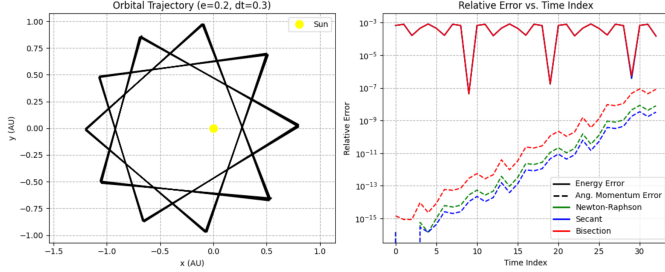


Fig. 4. Simulation results for $e = 0.2$ with an aggressive timestep ($\Delta t = 0.3$ years). The "star-shaped" aliasing in the left panel and the rapid secular growth in the right panel demonstrate the loss of symplectic integrity when the sampling rate is insufficient.

The right panel shows that the relative error in angular momentum is already starting to drift significantly, reaching 10^{-7} after only 30 time steps. Among the three implemented methods, the Bisection method consistently exhibits the highest relative error, then Newton-Raphson and Secant, respectively. This rapid secular growth suggests that when the timestep is large relative to the orbital period, the iterative solver struggles to find an accurate root for ΔE , leading to poor state updates. Even for a low-eccentricity orbit ($e = 0.2$), this highlights that the analytical strength of the f and g method is still bounded by the necessity of a reasonable temporal resolution to maintain Hamiltonian integrity.

D. Numerical Instability in High-Eccentricity Regimes

Figure 6 represents the stability threshold for the current numerical setup. At $e = 0.7$ and $\Delta t = 0.01$ years, the trajectory in the left panel appears visually stable and closed, showing that the f and g method can still resolve the orbital geometry reasonably well at this eccentricity. However, the error analysis reveals a clear secular drift in the constants of motion. The angular momentum error starts near the limit precision (10^{-16}) and grows exponentially over time, reaching 10^{-6} by the end of the simulation. Each step in the error plot corresponds to a periapsis passage, where the high-velocity gradient introduces small but consistent numerical inaccuracies. The system has not yet diverged, but the steady rise in relative error suggests that the simulation would eventually become unphysical over a longer temporal horizon.

Figure 7 illustrates the simulation for a highly elliptical orbit ($e = 0.8$) with the same timestep $\Delta t = 0.01$ years. Unlike the stable $e = 0.2$ case, the trajectory here exhibits orbital precession and eventual divergence, where the orbit

fails to close and instead spirals away from the Sun. This is a classic manifestation of numerical instability when the system's dynamics change faster than the solver can accurately resolve. All three methods (Newton-Raphson, Secant, and Bisection) fail at nearly the same rate. While the angular momentum errors begin near the limit precision (10^{-15}), they exhibit an exponential growth rate over time. Every 100 steps, the error jumps by orders of magnitude. By the 1000th step, the relative error has reached 10^0 error, which means the simulation has completely lost physical meaning. The spikes in the error plot align with the moment the body is closest to the Sun, the velocity is so high that the solver fails to find a root that satisfies the energy conservation shell.

The same terminal failure happens in Figure 8, under the extreme dynamical stress of $e = 0.9$. The trajectory exhibits a catastrophic divergence, the body is unphysically ejected from the system along a linear path. The error in energy angular momentum reaches a critical threshold near the 700th time index, where errors exceed 10^1 (1,000%). At such high eccentricity, the velocity gradient near periapsis is too steep for the fixed timestep of $\Delta t = 0.01$ years to resolve. The resulting "shocks" to the state vector at each close approach accumulate, demonstrating that adaptive timestepping or regularization is mandatory for simulating high-eccentricity regimes.

E. Impact of Mass Ratio on Relative Motion Dynamics

Figure 5 displays the system's response when the mass of the secondary body is increased to $m_1 = 0.9m_0$. While this represents a significant increase in the system's total mass, the visual impact on the orbital trajectory is negligible. This is because the f and g propagator is derived from the relative two-body equation, where the motion depends on the gravitational parameter $\mu = G(m_0 + m_1)$. In our canonical units, increasing m_1 primarily scales the orbital period and velocity, but it does not alter the fundamental elliptical geometry of the relative orbit unless the initial conditions are changed to compensate for the higher potential energy.

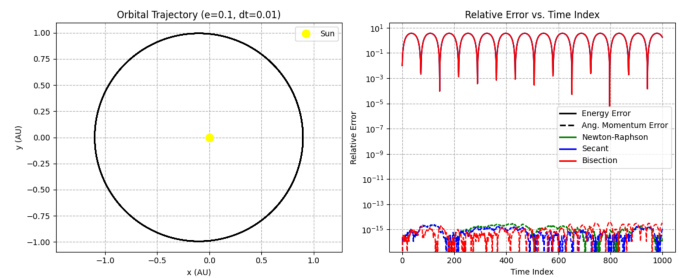


Fig. 5. System behavior under a significant mass ratio increase ($m_1 = 0.9m_0$). The geometric invariance of the relative trajectory confirms that the f and g formulation is primarily sensitive to the gravitational parameter μ rather than individual mass values.

The error analysis in the right panel shows that the energy error remains high, fluctuating near 10^0 for most solvers,

while the angular momentum error stays remarkably stable at the limit precision (10^{-15}). This indicates that while the system is physically more massive, the numerical challenges remain the same: the transcendental solver's tolerance is the primary bottleneck for energy conservation, whereas the angular momentum is more naturally conserved by the Lagrangian coefficient formulation regardless of the specific mass ratio.

F. Numerical Stability and Conservation Laws

The performance of the f and g propagator is fundamentally constrained by the convergence of the transcendental solvers and the resulting preservation of Hamiltonian invariants. In this study specifically, the Newton-Raphson, Secant, and Bisection methods were evaluated for their ability to maintain the constants of motion: Specific Orbital Energy (E) and Specific Relative Angular Momentum (l).

The Newton-Raphson method demonstrated superior efficiency in low-to-moderate eccentricity cases ($e \leq 0.5$), typically reaching the residual tolerance within 3 to 5 iterations due to its quadratic convergence. The Secant method provided a competitive alternative without requiring analytical derivatives, though it occasionally required additional iterations. In contrast, the Bisection method exhibited robust but strictly linear convergence; while it avoided divergence in high-eccentricity regimes, it often hit a higher precision floor than the gradient-based methods.

A primary metric for the propagator's accuracy is its adherence to conservation laws. Across all stable simulations, Angular Momentum consistently maintained a floor near the limit precision (10^{-16}), as the Lagrangian formulation naturally preserves the directional integrity of the orbital plane. However, energy conservation proved more sensitive, exhibiting fluctuations that peaked at periapsis. In high-eccentricity cases, these errors exhibited exponential secular drift, leading to unphysical orbital escape. This indicates that while the f and g method is analytically exact, maintaining symplectic integrity over long durations requires a precise balance between timestep resolution and a transcendental solver tolerance of at least 10^{-16} .

IV. CONCLUSION

The report demonstrates that the f and g Lagrangian coefficient propagator is a robust analytical tool for two-body dynamics, capable of preserving the orbital manifold even under sparse temporal sampling. While the method achieves the limit precision (10^{-16}) in angular momentum conservation for low-eccentricity orbits, it faces significant stability challenges in high-eccentricity regimes ($e \geq 0.8$), where steep velocity gradients at periapsis trigger exponential secular drift and unphysical divergence. Among the numerical solvers evaluated, the Newton-Raphson and Secant methods proved most efficient for standard orbits due to their superior convergence rates while the Bisection method is insufficient to maintain the desired energy precision. Ultimately, this report

demonstrates that while the f and g method is an exceptionally powerful tool for spacecraft and planetary trajectories, its long-term reliability is contingent upon a precise balance between solver tolerance and temporal resolution.

APPENDIX A NUMERICAL INSTABILITY IN HIGH-ECCENTRICITY REGIMES

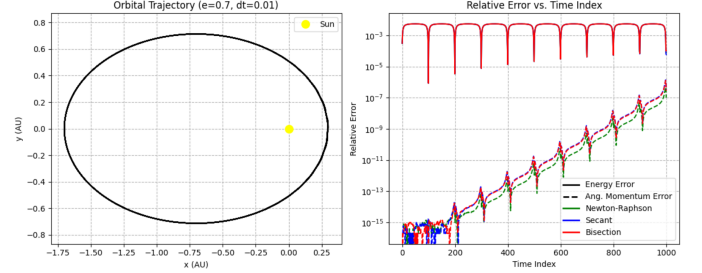


Fig. 6. Transition to instability at $e = 0.7$. Although the trajectory remains visually closed, the error analysis reveals the exponential secular drift in angular momentum following each periapsis passage.

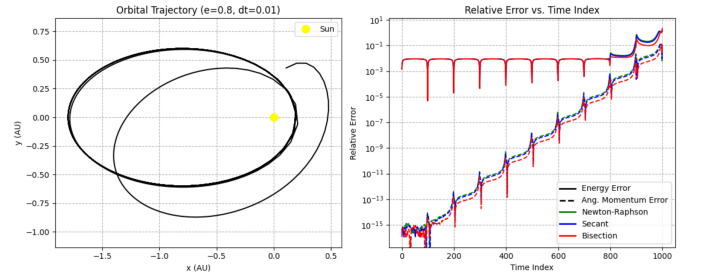


Fig. 7. Orbital divergence at $e = 0.8$. The accumulation of numerical errors at periapsis leads to orbital precession and a 100% relative error by the 1000th time index, signifying a total loss of physical validity.

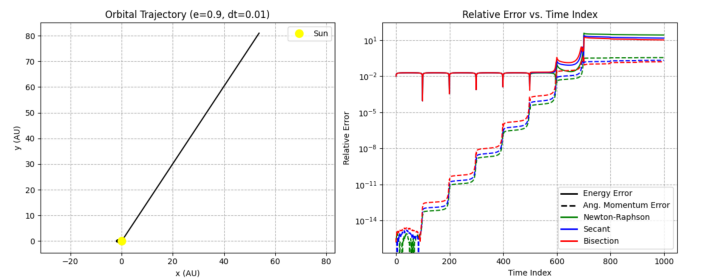


Fig. 8. Catastrophic failure and unphysical escape at $e = 0.9$.

ACKNOWLEDGMENT

I would like to express my gratitude to Asst. Prof. David M. Hernandez, the instructor of the Celestial Mechanics course, for providing the theoretical framework and project guidelines that shaped this study. I also acknowledge Dr. Karan Fernandes, Postdoctoral Researcher at the Center of Astronomy and Gravitation, for helping me develop a deeper understanding beyond the course lectures. Finally, I acknowledge the use of Google Gemini as an AI collaborative tool throughout the

completion of this project. Specifically, the AI was utilized in assisting in structural formatting and stylistic refinement of the manuscript.

REFERENCES

- [1] Battin, R. H. (1999). An Introduction to the Mathematics and Methods of Astrodynamics. AIAA Education Series.
- [2] Murray, C. D., & Dermott, S. F. (1999). Solar System Dynamics. Cambridge University Press.
- [3] Vallado, D. A. (2013). Fundamentals of Astrodynamics and Applications. Microcosm Press.
- [4] Goldstein, H., Poole, C., & Safko, J. (2002). Classical Mechanics. 3rd Edition, Addison-Wesley.
- [5] Danby, J. M. A. (1988). Fundamentals of Celestial Mechanics. Willmann-Bell.
- [6] Press, W. H., et al. (2007). Numerical Recipes: The Art of Scientific Computing. Cambridge University Press.